HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **computer vision models** for defect detection and anomaly identification on manufacturing lines, achieving a **92% accuracy rate** in identifying critical flaws.
- Engineered image processing pipelines using **Python** and **OpenCV** to enhance image quality and extract relevant features for downstream AI analysis, reducing false positives by **18%**.
- Designed and implemented rapid **proof-of-concepts** for clients, showcasing the capabilities of AI in identifying production bottlenecks and quality deviations.
- Built and managed diverse datasets, including image collection, labelling, and augmentation for **deep learning models**, ensuring sufficient data for robust training.
- Applied **convolutional neural networks** (CNNs) and transfer learning techniques to classify product types and detect specific object instances within industrial environments.
- Integrated vision solutions with **industrial cameras** and lighting systems, optimising parameters like exposure, gain, and frame rate to capture high-quality image data.
- Containerized computer vision applications using **Docker** and deployed them to **NVIDIA Jetson** edge devices for real-time inference on the factory floor.

SKILLS

Computer Vision & AI – OpenCV, PyTorch, TensorFlow, Keras, CNNs, ResNet, MobileNet, YOLO, U-Net, Classification, Object Detection, Defect Detection, Segmentation, Anomaly Detection, Feature Extraction, Video Analytics, Model Training, Model Evaluation, Image Augmentation, Transfer Learning

Programming & Scripting – Python, Bash Scripting

Imaging & Optics – Industrial Cameras, Lenses, Lighting, Field of View, Resolution, Depth of Field, Exposure, Gain, Frame Rate, Monochrome Imaging, Colour Imaging

Manufacturing & Integration – Industrial PCs, Edge Devices, PLCs, Robotics, SCADA, MES, Automation Systems

Deployment & Operations – Docker, NVIDIA Jetson, Linux, CI/CD, Git

PROJECTS

Manufacturing Defect Detection System

- Developed a **computer vision** pipeline to identify and classify defects on manufactured goods using **Python** and **PyTorch**, achieving **91% precision**.
- Trained a **U-Net model** on a dataset of 50,000 images to perform precise segmentation of defective areas, enabling automated quality control.
- Integrated the solution with an **industrial camera** and lighting setup, optimizing image acquisition for consistent performance across varied product batches.

Automated Measurement and Anomaly Detection

- Built a system using **OpenCV** and **scikit-learn** for automated measurement of critical component dimensions, reducing manual inspection time by **40%**.
- Implemented an **anomaly detection** algorithm to flag unusual patterns in process data and imagery, preventing the shipment of potentially faulty units.
- Deployed the application on an **edge device** for real-time analysis directly on the production line.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)